

SUNG KYU LIM

professor

基本信息

姓名: SUNG KYU LIM

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地区: United States

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

数字集成电路设计

EDA工具与电路验证

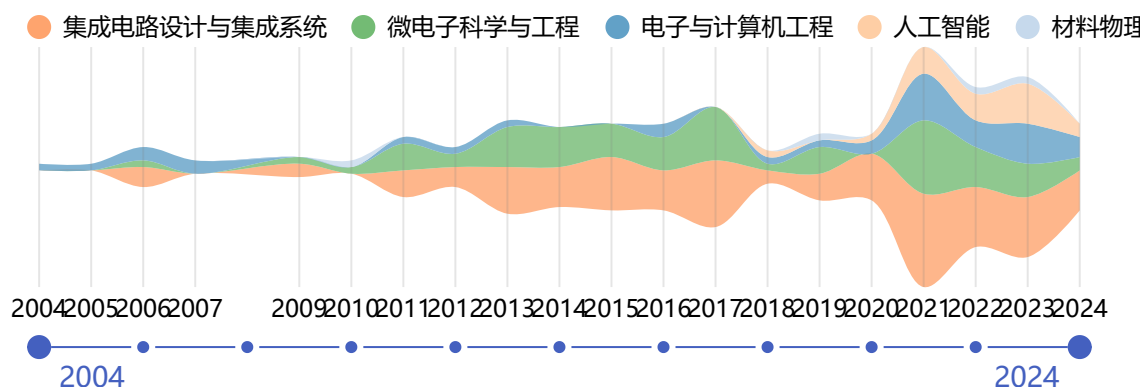
低功耗电路与能量管理

超大规模集成电路设计

个人简介

SUNG KYU LIM is a Professor at the School of Electrical and Computer Engineering, Georgia Institute of Technology, where he directs the GTCAD Laboratory. Born and raised in Seoul, Korea, he moved to Los Angeles at age 19 and earned his B.S., M.S., and Ph.D. in Computer Science from UCLA. His research focuses on physical design automation for VLSI circuits, 3D circuit/package layout automation, quantum circuit layout automation, micro-architecture design space exploration, and layout automation for reconfigurable circuits. He is also involved in interdisciplinary research through the Matter and Systems Institute.

近年发表的论文



Hier-3D: A Methodology for Physical Hierarchy Exploration of 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Bethur, Nesara Eranna; Agnesina, Anthony; Brunion, Moritz; Garcia-Ortiz, Alberto; Catthoor, Francky; Milojevic, Dragomir; Komalan, Manu; Cavalcante, Matheus; Riedel, Samuel; Benini, Luca; Lim, Sung Kyu

被引用次数: 0

2024

GAN-Place: Advancing Open Source Placers to Commercial-quality Using Generative Adversarial Networks and Transfer Learning

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Lu, Yi-Chen; Ren, Haoxing; Hsiao, Hao-Hsiang; Lim, Sung Kyu

被引用次数: 0

2024

3DNN-Xplorer: A Machine Learning Framework for Design Space Exploration of Heterogeneous 3-D DNN Accelerators

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Murali, Gauthaman; Park, Min Gyu; Lim, Sung Kyu

被引用次数: 0

2024

Heterogeneous Monolithic 3-D IC Designs: Challenges, EDA Solutions, and Power, Performance, Cost Tradeoffs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Pentapati, Sai; Lim, Sung-Kyu

被引用次数: 0

2024

A PPA Study for Heterogeneous 3-D IC Options: Monolithic, Hybrid Bonding, and Microbumping

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Jinwoo; Zhu, Lingjun; Torun, Hakki Mert; Swaminathan, Madhavan; Lim, Sung Kyu

被引用次数: 2

2024

Hetero-3D: Maximizing Performance and Power Delivery Benefits of Heterogeneous 3D ICs

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Zhu, Lingjun; Hu, Jiawei; Murali, Gauthaman; Lim, Sung Kyu

被引用次数: 0

2024

FastTuner: Transferable Physical Design Parameter Optimization using Fast Reinforcement Learning

PROCEEDINGS OF THE 2024 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2024

Hsiao, Hao-Hsiang; Lu, Yi-Chen; Vanna-Iampikul, Pruek; Lim, Sung Kyu

被引用次数: 3

2024

GNN-assisted Back-side Clock Routing Methodology for Advance Technologies

Design Automation Conference

Nesara Eranna Bethur; Pruek Vanna-Iampikul; Odysseas Zografos; Lingjun Zhu; Giuliano Sisto; Dragomir Milojevic; Alberto Garcia-Ortiz; Geert Hellings; Julien Ryckaert; Francky Catthoor; Sung Kyu Lim

被引用次数: 0

2024

ML-based Physical Design Parameter Optimization for 3D ICs: From Parameter Selection to Optimization

Design Automation Conference

Hao-Hsiang Hsiao; Pruek Vanna-Iampikul; Yi-Chen Lu; Sung Kyu Lim

被引用次数: 0

2024

Transferable Graph Neural Network-Based Delay-Fault Localization for Monolithic 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hung, Shao-Chun; Banerjee, Sanmitra; Chaudhuri, Arjun; Kim, Jinwoo; Lim, Sung Kyu; Chakrabarty, Krishnendu

被引用次数: 1

2023

Snap-3D: A Constrained Placement-Driven Physical Design Methodology for High Performance 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Vanna-Iampikul, Pruek; Shao, Chengjia; Lu, Yi-Chen; Pentapati, Sai; Heo, Yun; Choi, Jae-Seung; Lim, Sung Kyu

被引用次数: 1

2023

Parameter Optimization of VLSI Placement Through Deep Reinforcement Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Agnesina, Anthony; Chang, Kyungwook; Lim, Sung Kyu

被引用次数: 4

2023

Abisko: Deep codesign of an architecture for spiking neural networks using novel neuromorphic materials

INTERNATIONAL JOURNAL OF HIGH PERFORMANCE COMPUTING APPLICATIONS

Vetter, Jeffrey S.; Date, Prasanna; Fahim, Farah; Kulkarni, Shruti R.; Maksymovych, Petro; Talin, A. Alec; Tallada, Marc Gonzalez; Vanna-Iampikul, Pruek; Young, Aaron R.; Brooks, David; Cao, Yu; Gu-Yeon, Wei; Lim, Sung Kyu; Liu, Frank; Marinella, Matthew; Sumpter, Bobby; Miniskar, Narasinga Rao

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2023

A PPA Study of Reinforced Placement Parameter Autotuning: Pseudo-3D vs. True-3D Placers

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Murali, Gauthaman; Agnesina, Anthony; Lim, Sung Kyu

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2023

GNN-based Multi-bit Flip-flop Clustering and Post-clustering Design Optimization for Energy-efficient 3D ICs

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Vanna-lampikul, Pruek; Lu, Yi-Chen; Shim, Da Eun; Lim, Sung Kyu

被引用次数: 0

2023

ECO-GNN: Signoff Power Prediction Using Graph Neural Networks with Subgraph Approximation

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Lu, Yi-Chen; Nath, Siddhartha; Pentapati, Sai; Lim, Sung Kyu

被引用次数: 0

2023

On Continuing DNN Accelerator Architecture Scaling Using Tightly Coupled Compute-on-Memory 3-D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Murali, Gauthaman; Iyer, Aditya; Zhu, Lingjun; Tong, Jianming; Martinez, Francisco Munoz; Srinivasa, Srivatsa Rangachar; Karnik, Tanay; Krishna, Tushar; Lim, Sung Kyu

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2023

Built-In Self-Test of High-Density and Realistic ILV Layouts in Monolithic 3-D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chaudhuri, Arjun; Banerjee, Sanmitra; Kim, Jinwoo; Lim, Sung Kyu; Chakrabarty, Krishnendu

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2023

Power, Performance, Area, and Cost Analysis of Face-to-Face-Bonded 3-D ICs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Agnesina, Anthony; Brunion, Moritz; Kim, Jinwoo; Garcia-Ortiz, Alberto; Milojevic, Dragomir; Catthoor, Francky; Mirabelli, Gioele; Komalan, Manu; Lim, Sung Kyu

被引用次数: 1

2023

Ultra-Dense 3D Physical Design Unlocks New Architectural Design Points with Large Benefits

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Srimani, Tathagata; Radway, Robert M.; Kim, Jinwoo; Prabhu, Kartik; Rich, Dennis; Gilardi, Carlo; Raina, Priyanka; Shulaker, Max; Lim, Sung Kyu; Mitra, Subhasish

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2023

On Legalization of Die Bonding Bumps and Pads for 3D ICs

PROCEEDINGS OF THE 2023 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2023

Pentapati, Sai; Agnesina, Anthony; Brunion, Moritz; Huang, Yen-Hsiang; Lim, Sung Kyu

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2023

DREAM-GAN: Advancing DREAMPlace towards Commercial-Quality using Generative Adversarial Learning

PROCEEDINGS OF THE 2023 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2023

Lu, Yi-Chen; Ren, Haoxing; Hsiao, Hao-Hsiang; Lim, Sung Kyu

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2023

RL-CCD: Concurrent Clock and Data Optimization using Attention-Based Self-Supervised Reinforcement Learning

Design Automation Conference

Yi-Chen Lu; Wei-Ting Chan; Deyuan Guo; Sudipto Kundu; Vishal Khandelwal; Sung Kyu Lim

被引用次数: 0

2023

Glass Interposer Integration of Logic and Memory Chiplets: PPA and Power/Signal Integrity Benefits

Design Automation Conference

Pruek Vanna-lampikul; Lingjun Zhu; Serhat Erdogan; Mohanalingam Kathaperumal; Ravi Agarwal; Ram Gupta; Kevin Rinebold; Sung Kyu Lim

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2023

A Machine Learning-Powered Tier Partitioning Methodology for Monolithic 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lu, Yi-Chen; Pentapati, Sai; Zhu, Lingjun; Murali, Gauthaman; Samadi, Kambiz; Lim, Sung Kyu

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2022

A Clock Tree Prediction and Optimization Framework Using Generative Adversarial Learning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lu, Yi-Chen; Lee, Jeehyun; Agnesina, Anthony; Samadi, Kambiz; Lim, Sung Kyu

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2022

Design-Aware Partitioning-Based 3-D IC Design Flow With 2-D Commercial Tools

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chang, Kyungwook; Sinha, Saurabh; Cline, Brian; Yeric, Greg; Lim, Sung Kyu

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2022

Unsupervised Digit Recognition Using Cosine Similarity In A Neuromemristive Competitive Learning System

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Ku, Bon Woong; Schuman, Catherine D.; Adnan, Md Musabbir; Mintz, Tiffany M.; Pooser, Raphael; Hamilton, Kathleen E.; Rose, Garrett S.; Lim, Sung Kyu

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Built-in Self-Test and Fault Localization for Inter-Layer Vias in Monolithic 3D ICs

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Chaudhuri, Arjun; Banerjee, Sanmitra; Kim, Jinwoo; Park, Heechun; Ku, Bon Woong; Kannan, Sukeshwar; Chakrabarty, Krishnendu; Lim, Sung Kyu

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2022

A ReRAM Memory Compiler for Monolithic 3D Integrated Circuits in a Carbon Nanotube Process

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Lee, Edward; Kim, Daehyun; Kim, Jinwoo; Lim, Sung Kyu; Mukhopadhyay, Saibal

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2022

Design Automation and Test Solutions for Monolithic 3D ICs

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Zhu, Lingjun; Chaudhuri, Arjun; Banerjee, Sanmitra; Murali, Gauthaman; Vanna-Iampikul, Pruek; Chakrabarty, Krishnendu; Lim, Sung Kyu

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2022

A Compute-in-Memory Hardware Accelerator Design With Back-End-of-Line (BEOL) Transistor Based Reconfigurable Interconnect

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Luo, Yandong; Dutta, Sourav; Kaul, Ankit; Lim, Sung Kyu; Bakir, Muhannad; Datta, Suman; Yu, Shimeng

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2022

Metal Layer Sharing: A Routing Optimization Technique for Monolithic 3D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Pentapati, Sai; Lim, Sung Kyu

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2022

Performance, Power, and Area of Standard Cells in Sub 3 nm Node Using Buried Power Rail

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoon, Jun-Sik; Jeong, Jinsu; Lee, Seunghwan; Lee, Junjong; Lee, Sanguk; Baek, Rock-Hyun; Lim, Sung Kyu

被引用次数: 10

2022

Power Delivery Solutions and PPA Impacts in Micro-Bump and Hybrid-Bonding 3D ICs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Zhu, Lingjun; Jo, Chanmin; Lim, Sung Kyu

被引用次数: 4

2022

Antiferroelectric negative capacitance from a structural phase transition in zirconia

NATURE COMMUNICATIONS

Hoffmann, Michael; Wang, Zheng; Tasneem, Nujhat; Zubair, Ahmad; Ravindran, Prasanna Venkatesan; Tian, Mengkun; Gaskell, Anthony Arthur; Triyoso, Dina; Consiglio, Steven; Tapily, Kandabara; Clark, Robert; Hur, Jae; Pentapati, Sai Surya Kiran; Lim, Sung Kyu; Dopita, Milan; Yu, Shimeng; Chern, Winston; Kacher, Josh; Reyes-Lillo, Sebastian E.; Antoniadis, Dimitri; Ravichandran, Jayakanth; Slesazeck, Stefan; Mikolajick, Thomas; Khan, Asif Islam

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2022

ART-3D: Analytical 3D Placement with Reinforced Parameter Tuning for Monolithic 3D ICs

ISPD'22: PROCEEDINGS OF THE 2022 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN

Murali, Gauthaman; Shaji, Sandra Maria; Agnesina, Anthony; Luo, Guojie; Lim, Sung Kyu

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2022

Routing Layer Sharing: A New Opportunity for Routing Optimization in Monolithic 3D ICs

ISPD'22: PROCEEDINGS OF THE 2022 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN

Pentapati, Sai; Lim, Sung Kyu

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2022

Worst-Case Eye Analysis of High-Speed Channels Based on Bayesian Optimization

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Dolatsara, Majid Ahadi; Hejase, Jose Ale; Becker, Wiren Dale; Kim, Jinwoo; Lim, Sung Kyu; Swaminathan, Madhavan

被引用次数: 9

2021

Pseudo-3D Physical Design Flow for Monolithic 3D ICs: Comparisons and Enhancements

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Park, Heechun; Ku, Bon Woong; Chang, Kyungwook; Shim, Da Eun; Lim, Sung Kyu

被引用次数: 2

2021

Power Supply Noise-Aware At-Speed Delay Fault Testing of Monolithic 3-D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hung, Shao-Chun; Lu, Yi-Chen; Lim, Sung Kyu; Chakrabarty, Krishnendu

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2021

High-Performance Logic-on-Memory Monolithic 3-D IC Designs for Arm Cortex-A Processors

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhu, Lingjun; Bamberg, Lennart; Pentapati, Sai Surya Kiran; Chang, Kyungwook; Catthoor, Francky; Milojevic, Dragomir; Komalan, Manu; Cline, Brian; Sinha, Saurabh; Xu, Xiaoqing; Garcia-Ortiz, Alberto; Lim, Sung Kyu

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2021

Clock Delivery Network Design and Analysis for Interposer-Based 2.5-D Heterogeneous Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Murali, Gauthaman; Park, Heechun; Qin, Eric; Torun, Hakki Mert; Dolatsara, Majid Ahadi; Swaminathan, Madhavan; Krishna, Tushar; Lim, Sung Kyu

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2021

Heterogeneous Mixed-Signal Monolithic 3-D In-Memory Computing Using Resistive RAM

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Murali, Gauthaman; Sun, Xiaoyu; Yu, Shimeng; Lim, Sung Kyu

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2021

Chiplet/Interposer Co-Design for Power Delivery Network Optimization in Heterogeneous 2.5-D ICs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Kim, Jinwoo; Chekuri, Venkata Chaitanya Krishna; Rahman, Nael Mizanur; Dolatsara, Majid Ahadi; Torun, Hakki Mert; Swaminathan, Madhavan; Mukhopadhyay, Saibal; Lim, Sung Kyu

被引用次数: 16

2021

Monolithic 3D Inverter with Interface Charge: Parameter Extraction and Circuit Simulation

APPLIED SCIENCES-BASEL

Ahn, Tae Jun; Lim, Sung Kyu; Yu, Yun Seop

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2021

FAFNIR: Accelerating Sparse Gathering by Using Efficient Near-Memory Intelligent Reduction

IEEE International Symposium on High Performance Computer Architecture

Bahar Asgari; Ramyad Hadidi; Jiashen Cao; Da Eun Shim; Sung-Kyu Lim; Hyesoon Kim

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2021

Micro-bumping, Hybrid Bonding, or Monolithic? A PPA Study for Heterogeneous 3D IC Options

Design Automation Conference

Jinwoo Kim; Lingjun Zhu; Hakki Mert Torun; Madhavan Swaminathan; Sung Kyu Lim

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2021

RL-Sizer: VLSI Gate Sizing for Timing Optimization using Deep Reinforcement Learning

Design Automation Conference

Yi-Chen Lu; Siddhartha Nath; Vishal Khandelwal; Sung Kyu Lim

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2021

Heterogeneous Monolithic 3D ICs: EDA Solutions, and Power, Performance, Cost Tradeoffs

Design Automation Conference

Sai Surya Kiran Pentapati; Sung Kyu Lim

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2021

An SRAM Compiler for Monolithic-3-D Integrated Circuit With Carbon Nanotube Transistors

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Kim, Daehyun; Lee, Edward; Seo, Jamin; Kim, Jinwoo; Lim, Sung Kyu; Mukhopadhyay, Saibal

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2021

Heterogeneous 3D ICs: Current Status and Future Directions for Physical Design Technologies

PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)

Murali, Gauthaman; Lim, Sung Kyu

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2021

Monolithic 3D Compute-in-Memory Accelerator with BEOL Transistor based Reconfigurable Interconnect

2021 IEEE INTERNATIONAL ELECTRON DEVICES MEETING (IEDM)

Luo, Yandong; Dutta, Sourav; Kaul, Ankit; Lim, Sung-kyu; Bakir, Muhannad; Datta, Suman; Yu, Shimeng

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2021

Statistical Array Allocation and Partitioning for Compute In-Memory Fabrics

VLSI-SOC: DESIGN TRENDS, VLSI-SOC 2020

Crafton, Brian; Spetalnick, Samuel; Murali, Gauthaman; Krishna, Tushar; Lim, Sung-Kyu; Raychowdhury, Arijit

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RL-Sizer: VLSI Gate Sizing for Timing Optimization using Deep Reinforcement Learning

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Lu, Yi-Chen; Nath, Siddhartha; Khandelwal, Vishal; Lim, Sung Kyu

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2021

Heterogeneous Monolithic 3D ICs: EDA Solutions, and Power, Performance, Cost Tradeoffs

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Pentapati, Sai Surya Kiran; Lim, Sung Kyu

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2021

Micro-bumping, Hybrid Bonding, or Monolithic? A PPA Study for Heterogeneous 3D IC Options

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Kim, Jinwoo; Zhu, Lingjun; Torun, Hakki Mert; Swaminathan, Madhavan; Lim, Sung Kyu

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2021

ParaMitE: Mitigating Parasitic CNFETs in the Presence of Unetched CNTs

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Banerjee, Sanmitra; Chaudhuri, Arjun; Kim, Jinwoo; Murali, Gauthaman; Nelson, Mark; Lim, Sung Kyu; Chakrabarty, Krishnendu

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Security Closure of Physical Layouts ICCAD Special Session Paper

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Knechtel, Johann; Gopinath, Jayanth; Bhandari, Jitendra; Ashraf, Mohammed; Amrouch, Hussam; Borkar, Shekhar; Lim, Sung-Kyu; Sinanoglu, Ozgur; Karri, Ramesh

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Doomed Run Prediction in Physical Design by Exploiting Sequential Flow and Graph Learning

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Lu, Yi-Chen; Nath, Siddhartha; Khandelwal, Vishal; Lim, Sung Kyu

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Compact-2D: A Physical Design Methodology to Build Two-Tier Gate-Level 3-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ku, Bon Woong; Chang, Kyungwook; Lim, Sung Kyu

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2020

Heterogeneous 3D Integration for a RISC-V System With STT-MRAM

IEEE COMPUTER ARCHITECTURE LETTERS

Zhu, Lingjun; Bamberg, Lennart; Agnesina, Anthony; Catthoor, Francky; Milojevic, Dragomir; Komalan, Manu; Ryckaert, Julien; Garcia-Ortiz, Alberto; Lim, Sung Kyu

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2020

Improving FPGA-Based Logic Emulation Systems through Machine Learning

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Agnesina, Anthony; Lim, Sung Kyu; Lepercq, Etienne; Del Cid, Jose Escobedo

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2020

A COTS-Based Novel 3-D DRAM Memory Cube Architecture for Space Applications

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Agnesina, Anthony; Yamaguchi, James; Krutzik, Christian; Carson, John; Yang-Scharlotta, Jean; Lim, Sung Kyu

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2020

Cross-Domain Optimization of Ferroelectric Parameters for Negative Capacitance Transistors-Part I: Constant Supply Voltage

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pentapati, Sai; Perumal, Rakesh; Khandelwal, Sourabh; Hoffmann, Michael; Lim, Sung Kyu; Khan, Asif I.

被引用次数: 8

2020

Optimal Ferroelectric Parameters for Negative Capacitance Field-Effect Transistors Based on Full-Chip Implementations-Part II: Scaling of the Supply Voltage

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pentapati, Sai; Perumal, Rakesh; Khandelwal, Sourabh; Khan, Asif I.; Lim, Sung Kyu

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2020

Design Flow for Active Interposer-Based 2.5-D ICs and Study of RISC-V Architecture With Secure NoC

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Park, Heechun; Kim, Jinwoo; Chekuri, Venkata Chaitanya Krishna; Dolatsara, Majid Ahadi; Nabeel, Mohammed; Bojesomo, Alabi; Patnaik, Satwik; Sinanoglu, Ozgur; Swaminathan, Madhavan; Mukhopadhyay, Saibal; Knechtel, Johann; Lim, Sung Kyu

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2020

Automated I/O Library Generation for Interposer-Based System-in-Package Integration of Multiple Heterogeneous Dies

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Lee, Minah; Singh, Arvind; Torun, Hakki Mert; Kim, Jinwoo; Lim, Sung Kyu; Swaminathan, Madhavan; Mukhopadhyay, Saibal

被引用次数: 1	2020
Advances in Design and Test of Monolithic 3-D ICs	
IEEE DESIGN & TEST	
Chaudhuri, Arjun; Banerjee, Sanmitra; Park, Heechun; Kim, Jinwoo; Murali, Gauthaman; Lee, Edward; Kim, Daehyun; Lim, Sung Kyu; Mukhopadhyay, Saibal; Chakrabarty, Krishnendu	
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TP-GNN: A Graph Neural Network Framework for Tier Partitioning in Monolithic 3D ICs	
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Yi-Chen Lu; Sai Surya Kiran Pentapati; Lingjun Zhu; Kambiz Samadi; Sung Kyu Lim	
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Entropy Production-Based Full-Chip Fatigue Analysis: From Theory to Mobile Applications	
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Wang, Tianchen; Samal, Sandeep Kumar; Lim, Sung Kyu; Shi, Yiyu	
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A Logic-on-Memory Processor-System Design With Monolithic 3-D Technology	
IEEE MICRO	
Pentapati, Sai; Zhu, Lingjun; Bamberg, Lennart; Shim, Da Eun; Garcia-Ortiz, Alberto; Lim, Sung Kyu	
被引用次数: 10	2019
System-Level Power Delivery Network Analysis and Optimization for Monolithic 3-D ICs	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Chang, Kyungwook; Das, Shidhartha; Sinha, Saurabh; Cline, Brian; Yeric, Greg; Lim, Sung Kyu	
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Parameter Extraction and Power/Performance Analysis of Monolithic 3-D Inverter (M3INV)	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Ahn, Tae Jun; Perumal, Rakesh; Lim, Sung Kyu; Yu, Yun Seop	
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Electrical Coupling and Simulation of Monolithic 3D Logic Circuits and Static Random Access Memory	
MICROMACHINES	
Ahn, Tae Jun; Choi, Bum Ho; Lim, Sung Kyu; Yu, Yun Seop	
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Power, Performance, and Area Benefit of Monolithic 3D ICs for On-Chip Deep Neural Networks Targeting Speech Recognition	
ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS	
Chang, Kyungwook; Kadetotad, Deepak; Cao, Yu; Seo, Jae-Sun; Lim, Sung Kyu	
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Yarui Peng; Dusan Petranovic; Sung Kyu Lim

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学术合作



Shreepad Panth	合作次数：7
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佐治亚理工学院	
Lingjun Zhu	合作次数：5
佐治亚理工学院	
Anthony Agnesina	合作次数：4
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